Lakitu Flight Data Analysis

Stratospheric Balloon Flight - 28 July 2026

Detailed post-flight telemetry, payload, health, and TT&C report

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Primary window:	14:29:00–17:32:00 CEST
Primary records:	103,510
Window duration:	3.05 h
Report build:	30 July 2026

Lakitu project team

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1 Executive summary

This report analyses the coherent public on-board flight window from 14:29 to 17:32 CEST, containing 103,510 rows sampled at a nominal 10 Hz. It also uses the 186,918-row merged archive for provenance and continuity context and the two ground-station CSV sessions for received-link statistics. The numerical summary, compact plotting tables, graph PDFs, and this report are all generated from the same reproducible analysis script.

Mission outcome. The platform reached a maximum GNSS altitude of 38.48 km at 16:15:11 CEST. The mean measured ascent rate from the Ascent transition to apex was 6.564 m/s; the maximum measured downward speed was 95.23 m/s. The GNSS launch-to-landing displacement was 37.499 km.

Primary performance discrepancy. At GNSS apex, the relative barometric altitude was 28.873 km, leaving a 9.607 km disagreement. This is too large to treat as ordinary sensor noise and requires calibration and atmospheric-model review.

Primary anomalies. The Coral link produced 15 timeout intervals and 1,620 invalid telemetry rows (1.565%). A power-on reset occurred at 16:51:39 CEST during Landing. No watchdog reset or non-Coral equipment fault was recorded.

The flight-state machine transitioned Standby → Launch → Ascent → Descent → Landing. It did not enter Cruise. This is consistent with the recorded post-apex descent condition being accepted before a 30-second cruise condition could be established, but it should be checked against the intended mission-state requirements.

2 Pre-flight preparation

The released dataset does not include a signed field checklist, photographs of the final integration state, or ground-station configuration records. The pre-flight assessment below therefore distinguishes readiness demonstrated by telemetry from preparation items that cannot be independently verified.

- **On-board readiness.** During the public Standby interval, GPS, IMU, barometer, battery, SD, and LoRa remained enabled and no corresponding equipment fault was recorded. The battery rail was centred on 3.856 V, and logging continued into Launch without a boot change.
- **Navigation readiness.** The first 3D GNSS fix was acquired at 14:37:35 CEST, approximately 81 seconds before the Standby-to-Launch transition at 14:38:56 CEST. This provided a valid position and altitude reference before release.
- **Ground-station reference.** For session 1, the station position is taken as the CubeSat's initial valid GNSS position, consistent with the co-located field setup. This assumption is sufficient for the range reconstruction but is not a substitute for a logged station GNSS track.
- **Configuration records.** The archive does not prove that antenna attachment, connector retention, payload communications, reset-cause capture, or end-to-end packet decoding were checked immediately before flight. These items should be explicit, signed checklist gates for the next campaign.

Readiness assessment. The telemetry shows a stable on-board Standby period and a valid GNSS solution before release. The main evidential gap is operational configuration traceability rather than an observed pre-launch subsystem fault.

3 Data scope and provenance

3.1 Analysed sources

Table 1: Source files used in this report.

Source	Role	Records
`flight_data_2026-07-28_1429-1732_CEST.csv`	Primary, coherent public flight-analysis window	103,510
`flight_data.csv`	Complete merged internal-memory archive; used for continuity and provenance checks	186,918
`telemetry_20260728_132651.csv` and `telemetry_20260728_165643.csv`	Ground-station decoded LoRa receptions and link measurements	622
`flight_data_audit.json`	Archive inventory, frame correlation, and source-file ledger	–
`flight_anomaly_report.md` and `data_gaps.md`	Independent anomaly and gap reviews used for reconciliation	–

The primary CSV is intentionally a time-window cut, not the whole internal memory image. The full audit reports 340 CSV files, 1,104 RAW Coral frames, 1,104 converted PNGs, and 1,101 RAW frames with a successful matching CSV Coral record. Photographs and RAW image content are outside this report's scope.

3.2 Time basis

The firmware stores GNSS time as `HHMMSS` without a calendar date. The date 2026-07-28 therefore comes from archive provenance, not from a GNSS date field. CEST is UTC+2 for this flight, so the public window corresponds to 12:29:00–15:32:00 UTC.

Within each run of identical one-second GNSS timestamps, this analysis uses the STM32 HAL tick to interpolate a sub-second plotting coordinate. Interpolation is restarted after an OBC boot because `record_timestamp_ms` resets. Displayed event times retain the authoritative GNSS second. This method avoids treating the HAL tick as continuous across the boot-5 to boot-6 reset.

3.3 Units and interpretation

Table 2: Principal field transformations.

CSV field	Transformation	Report unit
gps_lat_e7, gps_lon_e7	divide by 10^7	degrees
gps_alt_cm, baro_alt_cm	divide by 100	m
gps_speed_cms, gps_vel_down_cms	divide by 100; vertical plots negate down velocity	m/s
imu_accel_*_mg	divide by 1,000	g
imu_gyro_*_mdps	divide by 1,000	deg/s
baro_pressure_pa	divide by 100	hPa
baro_temp_centi_c	divide by 100	°C
batt_voltage_mv	divide by 1,000	V
coral_block_hex[4..5]	little-endian Q16 divided by 65,535	cloud fraction

The barometric altitude is relative to the Standby ground baseline; it is not an independent geodetic altitude. The MS5607 temperature channel is a sensor temperature used in pressure compensation and must not be presented as a calibrated ambient-air temperature.

4 Flight timeline and state-machine behaviour

Table 3: Recorded flight-state transitions. Positive vertical-down velocity denotes descent.

CEST	From	To	Elapsed	GNSS alt. (m)	Baro alt. (m)	v_D (m/s)
14:38:56	Standby	Launch	9:56	580	10.50	−3.82
14:39:08	Launch	Ascent	10:08	652	65.82	−7.55
16:15:22	Ascent	Descent	1:46:22	37,935	28,668.28	80.50
16:51:18	Descent	Landing	2:22:18	753	170.28	5.18

The state durations in the public window were 9.95 min in Standby, 11.2 s in Launch, 96.23 min in Ascent, 35.95 min in Descent, and 40.69 min in Landing. Cruise duration was zero. State persistence was effective: after the power-on reset, the first boot-6 record remained in Landing instead of reverting to Standby.

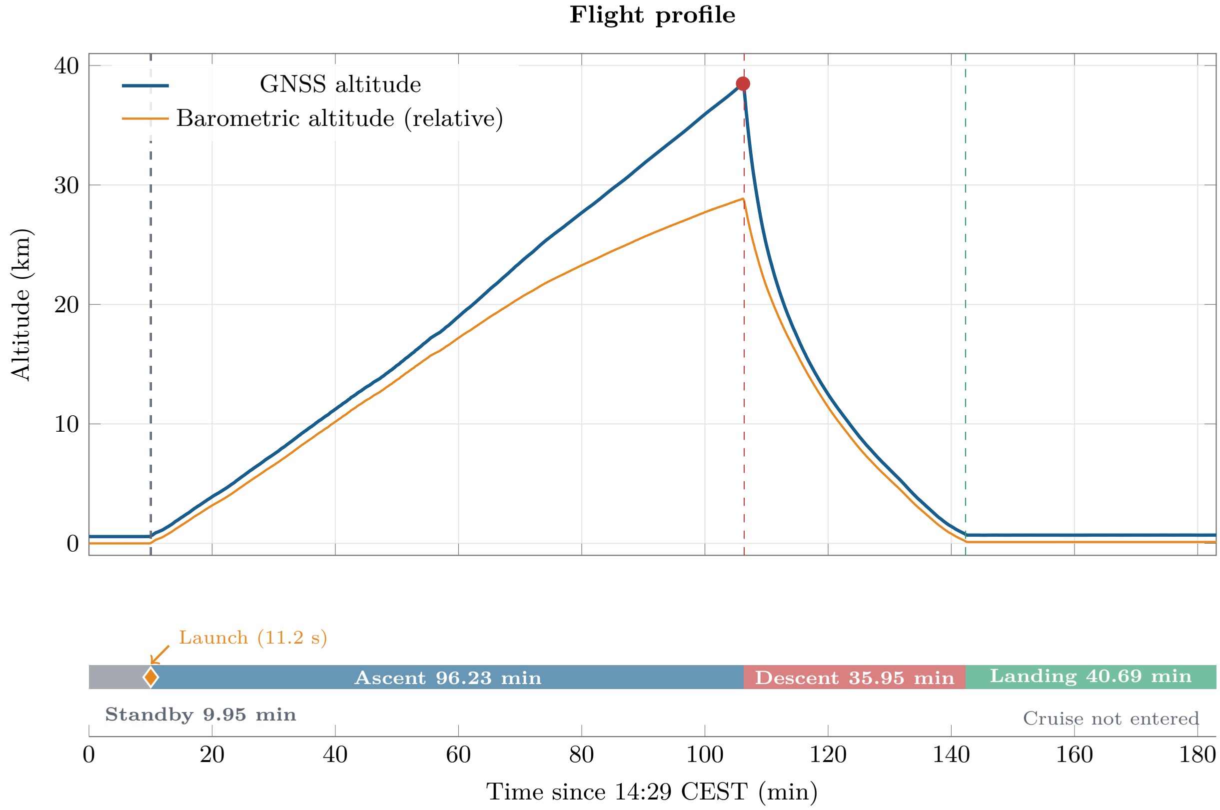

Figure 1: Flight profile.

Figure 1 plots five-second medians of GNSS and relative barometric altitude. Vertical lines mark the recorded phase transitions, and each lower-band rectangle begins and ends at the corresponding transition time. The duration is printed on or beside each band. The 11.2-second Launch state is identified by a diamond because its true interval is too narrow for a legible label at the full-flight scale; Cruise was not entered. Extrema and event values reported in the text and tables are calculated from the full-rate rows.

The GNSS profile is consistent with a sustained balloon ascent followed by a sharp burst/descent transition. The Ascent-to-Descent change occurs roughly 11 seconds after the maximum GNSS altitude record because the descent rule is debounced. A distinct float interval is not visible in the state record.

5 Altitude, velocity, and inertial dynamics

5.1 Performance metrics

Table 4: Flight-performance summary from the public on-board window.

Metric	Result
Maximum GNSS altitude	38.48 km at 16:15:11 CEST
Maximum relative barometric altitude	28.881 km
GNSS minus barometric altitude at GNSS apex	9.607 km
Mean ascent rate, Ascent transition to apex	6.564 m/s
Mean descent rate, apex to Landing transition	17.403 m/s
Maximum measured upward speed	16.14 m/s
Maximum measured downward speed	95.23 m/s
Maximum horizontal ground speed	38.89 m/s
Maximum acceleration magnitude	4.675 g at 14:35:24 CEST

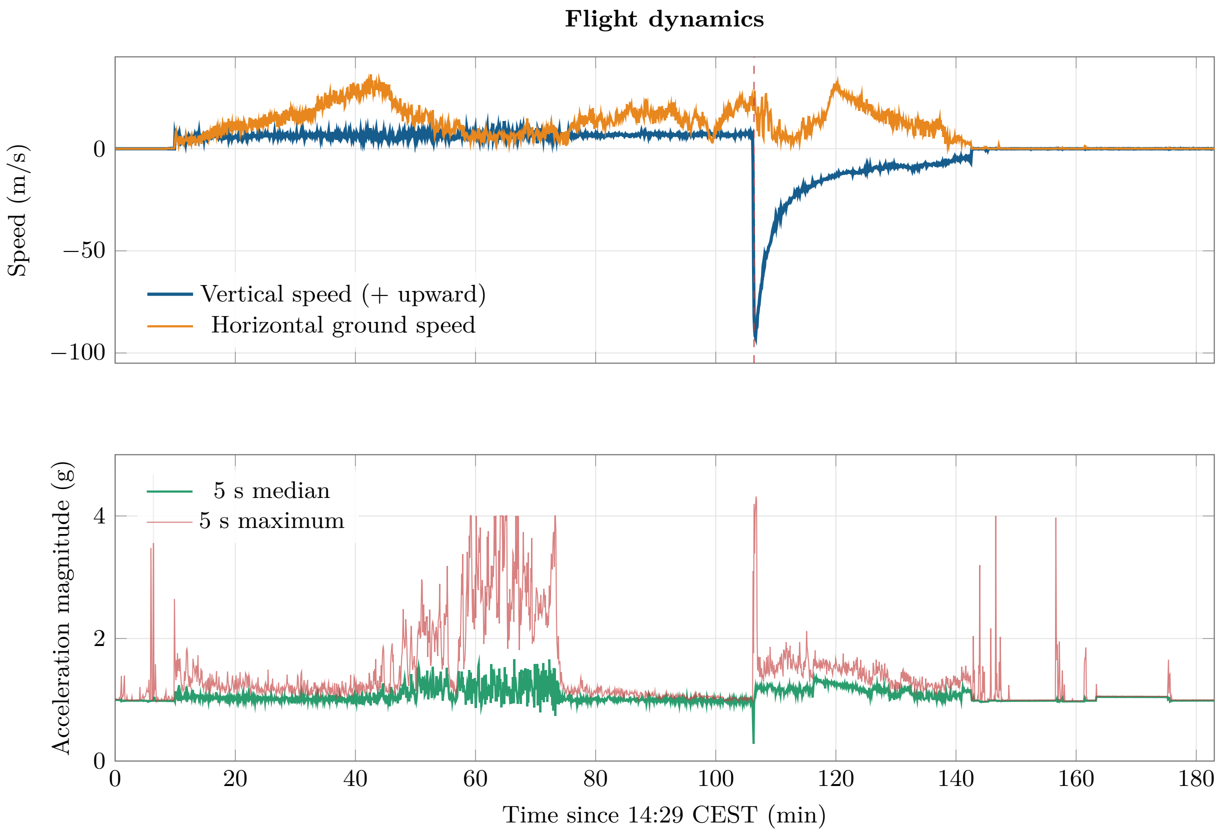

Figure 2: Flight dynamics.

Figure 2 compares GNSS vertical speed, horizontal ground speed, and IMU acceleration magnitude. Positive vertical speed is upward. The red acceleration trace retains the maximum value in each five-second plotting bin so short peaks remain visible beside the median trend.

The maximum 4.675 g magnitude occurred at 14:35:24, before the 14:38:56 Launch transition, with components (1.548, −2.045, −3.909) g. It is therefore more likely associated with pre-launch handling or vehicle dynamics than with balloon burst. The largest measured descent speed, 95.23 m/s, occurred immediately after the apex transition and then relaxed as the descent stabilised.

The 9.607 km altitude disagreement at apex is systematic rather than a single sample spike: the two profiles diverge progressively during ascent and reconverge during descent. Plausible contributors include the ground-relative barometric reference, an inappropriate fixed atmosphere model, pressure-sensor thermal behaviour, or model error in the very low pressure regime. Resolving the cause requires recalculating altitude from raw pressure with the intended reference pressure and comparing against GNSS altitude over pressure bands.

6 Atmospheric measurements

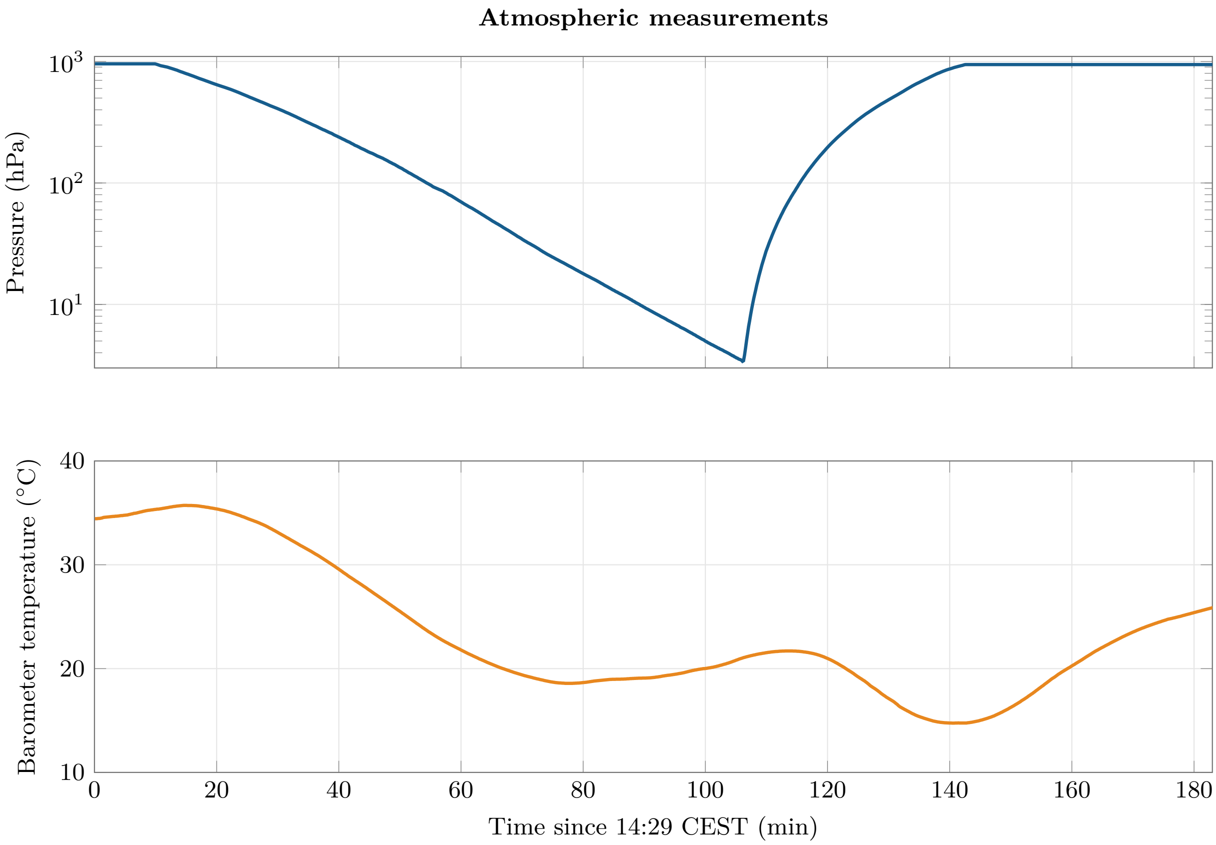

Figure 3: Atmospheric measurements.

Figure 3 uses ten-second medians and a logarithmic pressure axis to show the large pressure range without obscuring the surface portions of the flight. The MS5607 temperature trace is a sensor measurement used for pressure compensation, not a direct ambient-air measurement.

Pressure decreased from roughly 957 hPa near the start of the window to a minimum of 3.37 hPa near apex, then returned toward the surface value. The logarithmic profile is smooth and physically consistent with the gross altitude evolution; the main concern is the conversion from pressure to reported relative altitude rather than continuity of the pressure measurement.

The reported barometer temperature ranged from 14.74 to 35.72 °C. The channel changes smoothly, but its absolute values should be interpreted as device temperature within the enclosure. It cannot support an atmospheric lapse-rate conclusion without a separately calibrated external temperature sensor and a thermal-lag model.

7 GNSS trajectory

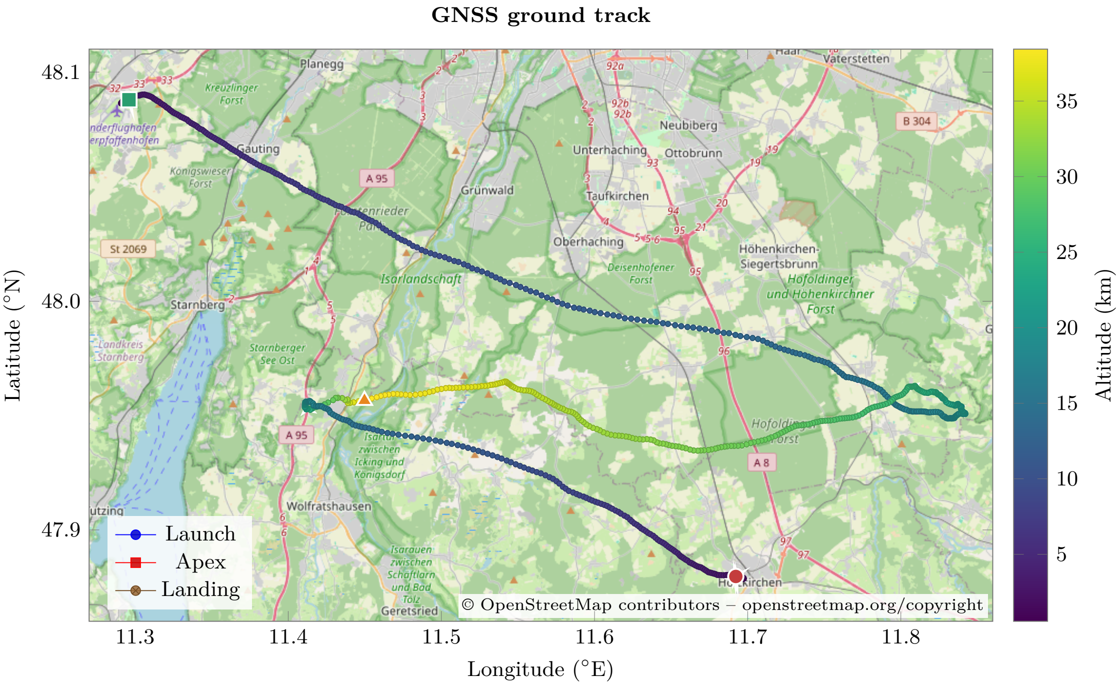

Figure 4: GNSS ground track.

Figure 4 plots ten-second median GNSS positions over an OpenStreetMap background in Web Mercator projection. Marker colour represents GNSS altitude, while launch, apex, and landing are identified separately. Map data: © OpenStreetMap contributors, https://www.openstreetmap.org/copyright.

Table 5: GNSS event-position estimates.

Event	Latitude (°N)	Longitude (°E)
Launch	48.0879328	11.2960224
Apex	47.9566112	11.4497968
Landing transition	47.879632	11.692252

The great-circle launch-to-landing displacement is 37.499 km; the maximum range from the launch position during the window is 43.389 km. These are point-to-point geodesic distances, not accumulated path length, so they are not inflated by small GNSS position jitter.

The first 3D fix appears at 14:37:35 CEST, 8.59 minutes after the start of the public window and before Launch. Prior coordinates in the CSV are carried values without a usable 3D solution and are excluded from the track and distance calculations.

8 Electrical power and IMU observations

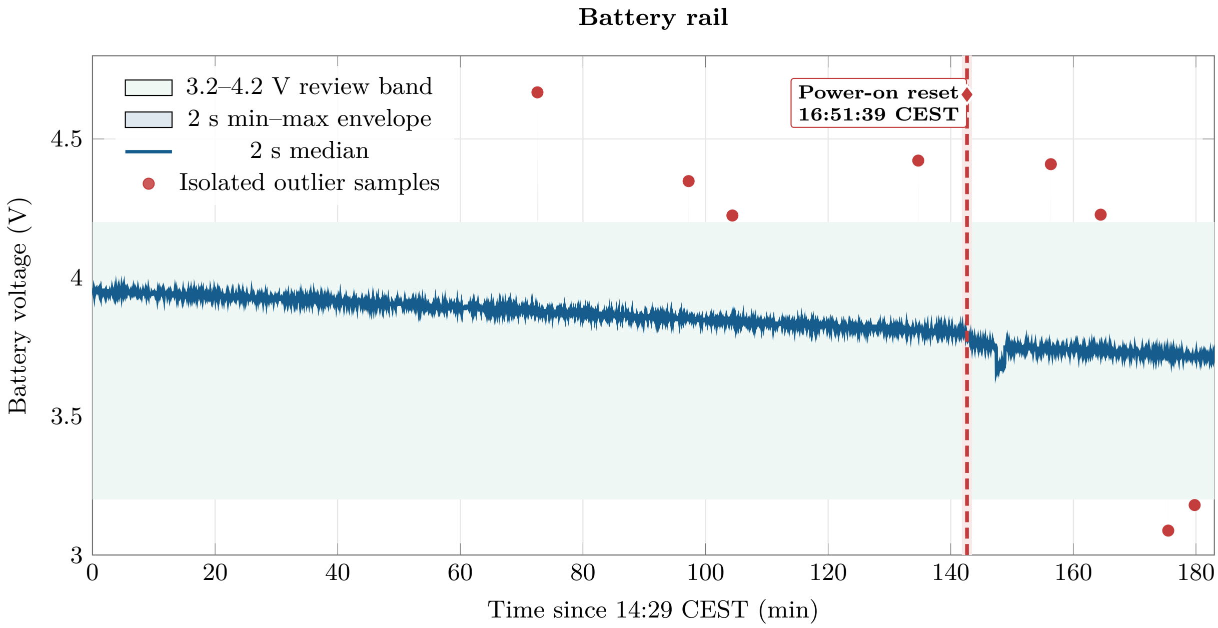

Figure 5: Battery rail.

Figure 5 shows the battery trend, two-second extrema, and isolated samples outside the 3.2–4.2 V review band. The power-on reset at 16:51:39 CEST is marked by a red band, dashed line, diamond, and callout. The green band is an analysis review threshold rather than a firmware-declared validity limit.

The battery median was 3.856 V, with a 5th–95th percentile range of 3.718–3.945 V. The full sample range was 3.088–4.668 V. Eight isolated rows fell outside the 3.2–4.2 V review band. Each extreme is short relative to the sampling cadence and the surrounding trend remains near the nominal rail, so the evidence favours ADC or switching-transient glitches over sustained overvoltage or undervoltage.

The voltage trend gradually declined through flight and stepped downward after the reboot, but the recorded battery-valid flag remained asserted throughout. Because the reset cause is explicitly power-on rather than watchdog, the power path, connector integrity, regulator undervoltage lockout, reset-cause capture, and ADC/reference behaviour should be examined together.

9 Coral payload results and link reliability

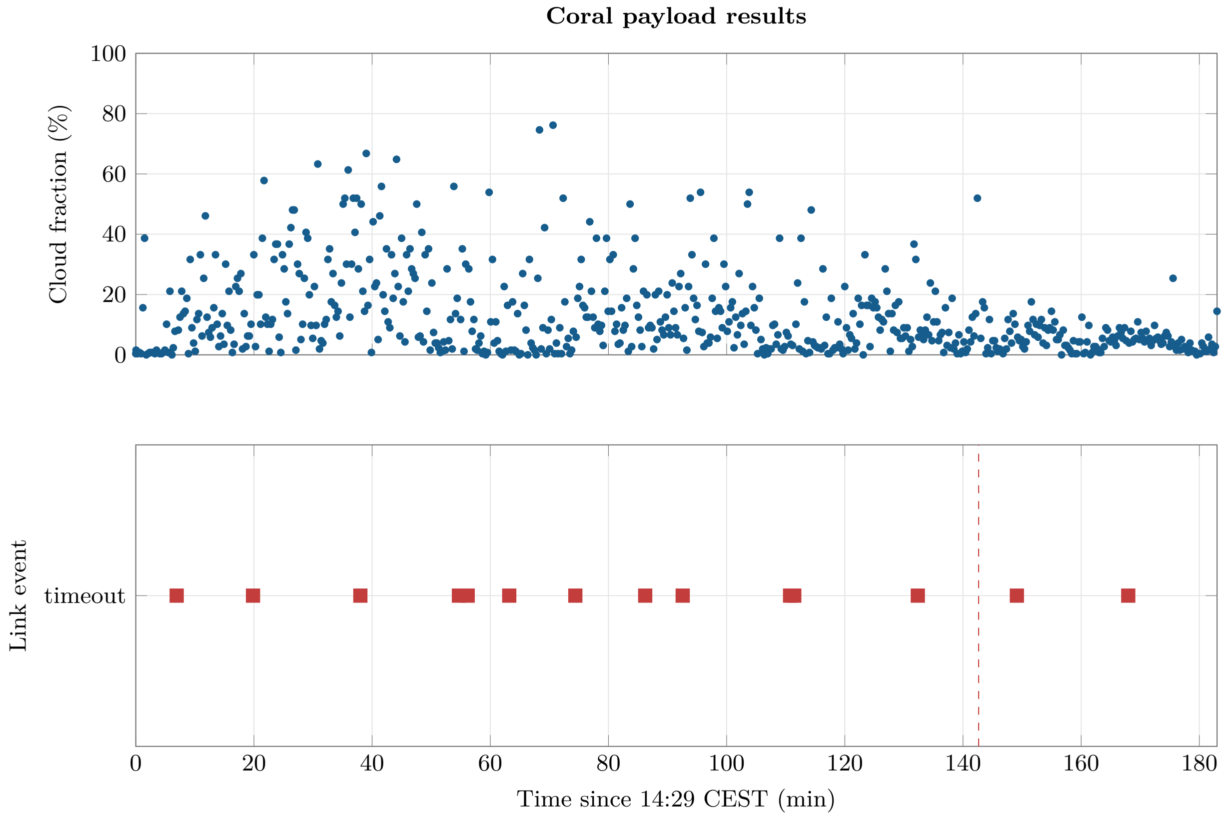

Figure 6: Coral payload results.

Figure 6 includes one point for each unique CRC-valid Coral cloud-fraction result and one marker for each unique timeout record. Repeated 10 Hz copies of the same 16-byte receipt block are removed, and the dashed line marks the OBC reboot.

The public window contains 630 unique successful Coral results. Their mean cloud fraction is 12.704%, the median is 8.006%, and the 90th percentile is 33.202%. Approximately 1.905% of successful results are exactly zero. These are model outputs; without human-labelled imagery they describe the payload response distribution, not independently verified cloud-cover truth.

The Coral channel was invalid for 1,620 rows (1.565%) and carried timeout status for 1,552 rows. The equipment fault bit shows 15 distinct intervals, totalling approximately 97.054 seconds of telemetry-tick duration; the longest was 21.636 seconds during post-reset recovery. No Coral CRC-error or Coral SD-error status was recorded.

Five in-window sequence omissions are adjacent to timeout intervals: 454→456, 620→622, 649→651, 729→731, and 816→818. The firmware stores a RAW frame only after a CRC-valid reception and deletes failed or partial frames. Accordingly, the sequence gaps indicate discarded receptions, not proven corruption of successfully stored image files.

10 TT&C downlink performance

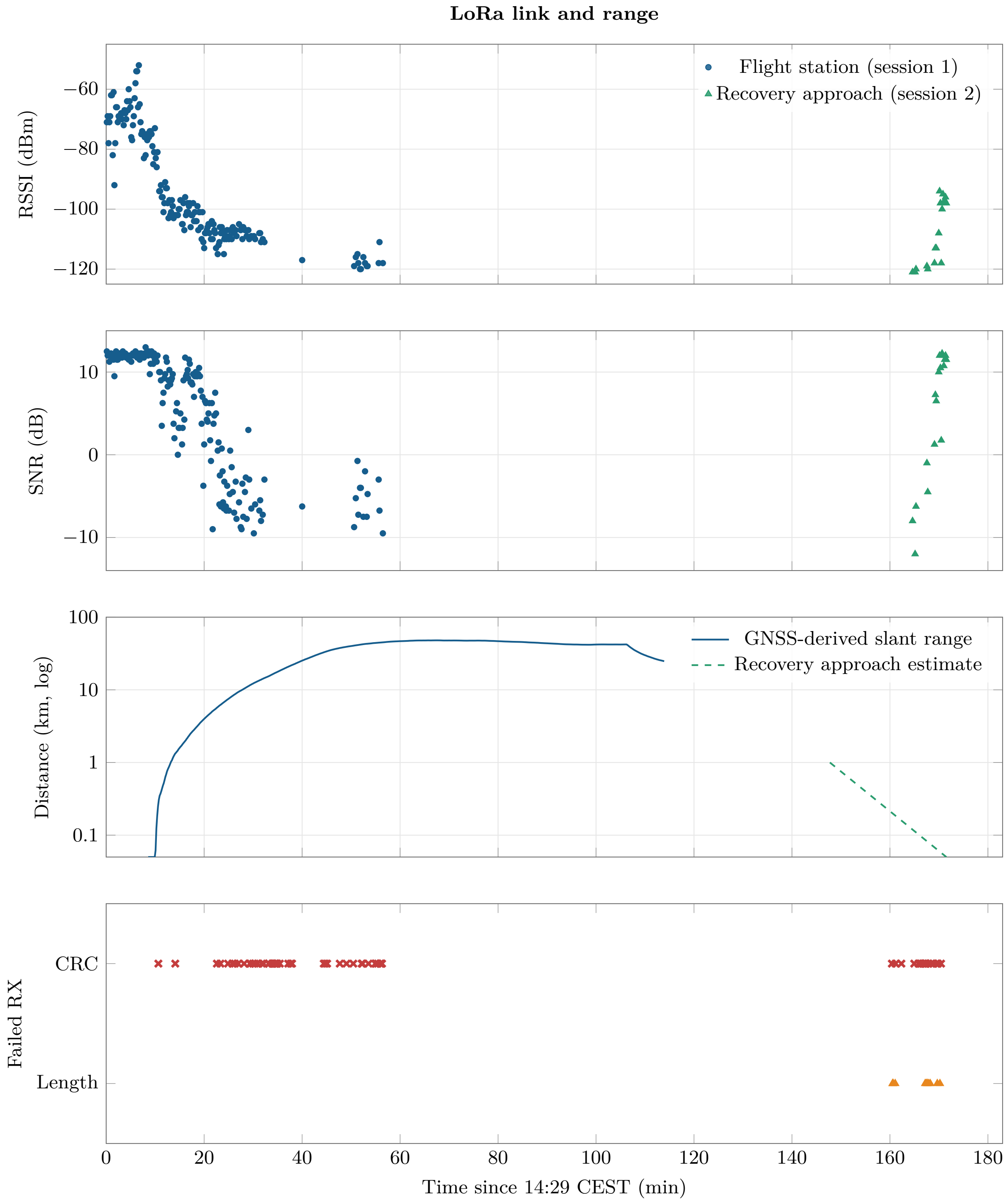

Figure 7: LoRa link and range.

Figure 7 aligns RSSI, SNR, ground-station distance, and failed receptions for the two LoRa sessions. The failure strip uses red crosses for 62 CRC failures and orange triangles for 12 unexpected-length receptions. Blank RSSI or SNR spans contain no valid decoded packet and are not interpolated.

Within the public flight window, the ground logs contain 200 valid decoded packets from 14:29:08 to 17:20:31 CEST and 62 LoRa CRC failures. The CRC failures represent 23.66% of the 200 valid plus 62 CRC-failed telemetry reception rows. A further 12 in-window receptions had unexpected lengths and never entered the decoded telemetry table. Across the complete ground-station sessions, including the period before the public flight window, the corresponding counts are 64 CRC failures and 12 unexpected-length packets. The valid packet sequence fields report 180 packets lost since the previous reception and 0 duplicates. Median RSSI was -101.5 dBm, spanning -121 to -52 dBm; median SNR was 8.63 dB, spanning -12 to 13 dB.

For session 1, the ground station is modelled at the CubeSat's initial valid GNSS position, consistent with the field setup, and held fixed. The resulting 3D slant range includes both horizontal separation and altitude difference. The last valid reception occurred at approximately 45.06 km slant range; by the later receiver stop the CubeSat was approximately 24.78 km away. This supports range as an important contributor to the observed progression from strong RSSI and positive SNR toward roughly −120 dBm and negative SNR, although it does not isolate propagation, orientation, or antenna effects.

The recovery station began session 2 at 16:56:44 CEST while the CubeSat was stationary. No station GNSS track was logged, so the dashed distance line is explicitly an estimate based on the operator account: the station began about 1 km away and approached the CubeSat, with co-location assigned to the final valid packet at 17:20:31 CEST. The renewed valid receptions and improving RSSI/SNR are consistent with the shrinking range, but the distance curve must not be interpreted as a measured trajectory. The logarithmic distance panel uses 0.05 km as a plotting floor at co-location.

The main antenna was detached during this transmission configuration and the radio radiated from the remaining feed cable. That non-standard radiator likely had a poorly controlled impedance and radiation pattern and may have reduced or made the effective radiated power strongly orientation-dependent. It is therefore a plausible contributor to the CRC errors, unexpected packet lengths, and other decode failures, together with range and propagation geometry; the logs cannot quantify its causal share. The long blank region must also be interpreted conservatively: these files describe what this ground station received, not every radio transmission. The on-board `scv_lora_timeout_count` and `scv_lora_tx_fault_counter` remained zero, so the data do not show an OBC-declared LoRa hardware fault.

11 Recovery

The state machine entered Landing at 16:51:18 CEST. A power-on reset was then observed at 16:51:39 CEST, but the first boot-6 record retained the Landing state and the persisted mission time. This allowed post-landing operation to continue without the software reverting to Standby. The cause of the interruption remains unresolved and should not be treated as a nominal recovery event.

The second ground-station session began at 16:56:44 CEST with the station approximately 1 km from the stationary CubeSat. The station approached the recovery point, and the range comparison assigns co-location to the final valid packet at 17:20:31 CEST. Valid packets reappeared and link quality improved as the estimated distance decreased, supporting successful close-range reacquisition.

The main LoRa antenna was detached during this configuration, so the radio was transmitting through the remaining feed cable. The cable was an uncontrolled radiator, and its impedance, radiation pattern, and orientation were not known. The recovery link therefore demonstrated that packets could still be received at close range, but it was not a representative end-to-end validation of the nominal antenna system. The lack of a logged station GNSS track also means that the plotted recovery distance is an operational estimate rather than a measurement.

Recovery assessment. The CubeSat remained operational after landing and was reacquired during the station approach, but the power-on reset, detached antenna, and unlogged station trajectory limit the conclusions that can be drawn from the recovery link.

12 System health, FDIR, and reboot

All records carry `scv_equipment_enabled`=127, meaning GPS, IMU, barometer, Coral, SD, LoRa, and EPS ADC remained enabled. The only equipment fault values are 0 and 8; bit 8 is the Coral fault. No row carries a GPS, IMU, barometer, SD, LoRa, EPS, or CDH fault. The I2C recovery state is always zero.

The maxima of the recorded GPS, IMU, barometer, LoRa, LoRa-TX, SD, and watchdog counters are all zero. GPS, IMU, barometer, and battery validity flags are each asserted for 100% of the primary rows. Coral is the sole channel with a material validity interruption.

The last boot-5 row is followed by boot 6 after an observed GNSS-time gap of 12.587 seconds, with the first new record at 16:51:39 CEST. `scv_reset_reason`=1 denotes a power-on reset. The watchdog reset counter remains zero, and the persisted phase and mission elapsed time resume in Landing. This demonstrates successful state persistence but leaves the physical source of the power interruption unresolved.

13 Data quality and limitations

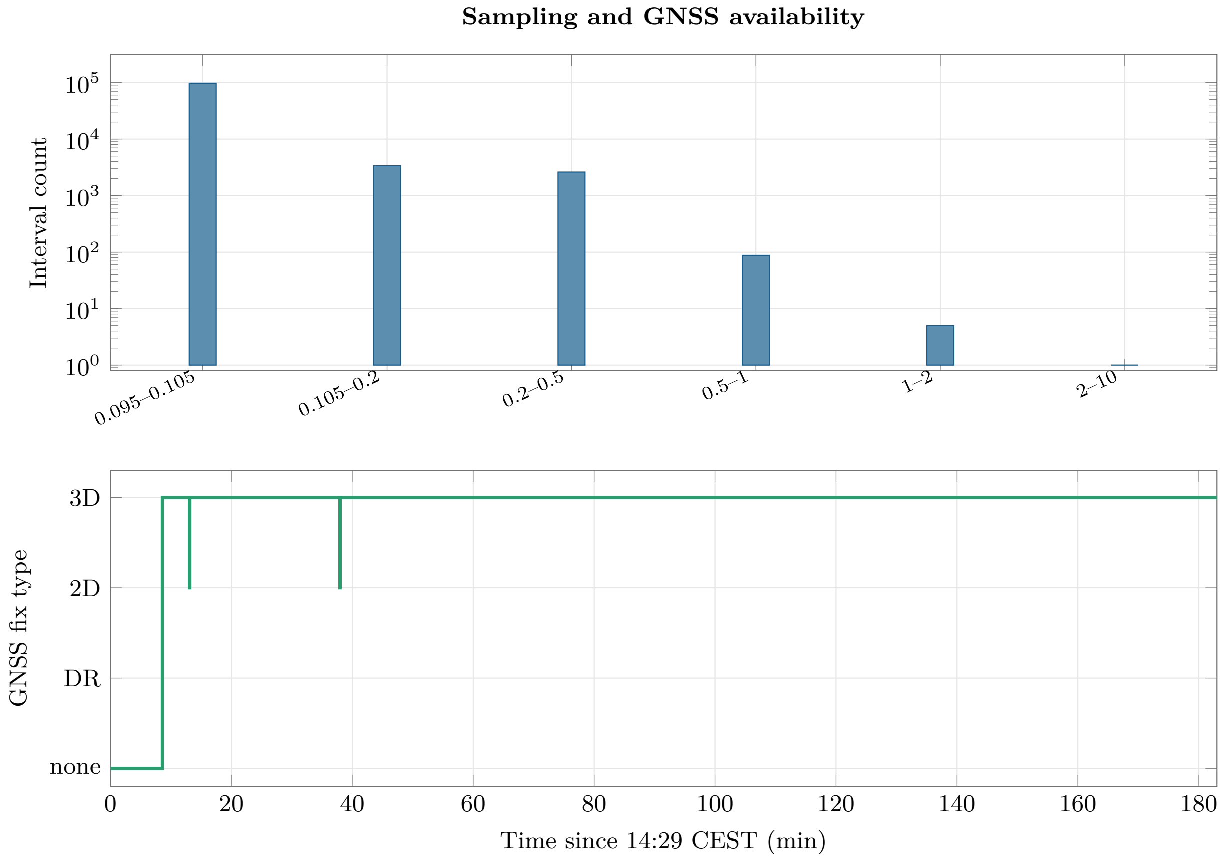

Figure 8: Sampling and GNSS availability.

Figure 8 combines the within-boot HAL-tick interval histogram with GNSS fix type over the mission timeline. The histogram excludes the boot boundary so the reset gap is not misclassified as a sampling stall.

The median within-boot sample interval is 100 ms and the 95th percentile is 109 ms. Six intervals are at least one second: five genuine stalls between 1.0 and 1.299 seconds plus one 8.673-second artefact caused by the public-cut filter. The full archive contains 81 valid rows in that interval, but their `gps_utc_time` is zero and they were excluded when the public file was selected by non-zero GNSS clock. No archive CSV-file sequence gap or within-boot gap above two seconds is reported by the full audit.

GNSS transport validity is 100%, while a 3D navigation fix is present for 98,643 rows (95.298%). The distinction is important: `gps_valid`=1 means the receiver data are fresh and accepted; it does not mean a usable 3D solution. The initial no-fix interval ends at 14:37:35, and three later 2D/no-fix interruptions are brief.

The following limitations govern interpretation:

- The GNSS calendar date is an archive assumption; only time of day is stored on board.
- The public CSV omits the 81 GNSS-time-zero rows; continuity conclusions use the full archive audit.
- Barometric altitude is relative and model-dependent. Raw pressure is the more direct observable.
- Barometer temperature is an internal sensor measurement, not validated ambient temperature.
- Cloud fractions have no manual ground-truth labels in the released dataset.
- Ground-station link logs are receive-side samples; missing receptions do not identify a unique

failure cause.
- Session 2 ground-station distance is an operator-informed linear estimate because no station GNSS track was recorded.
- Five- or ten-second medians are used only for graph readability. Summary extrema use full-rate rows.

14 Lessons learnt

1. **Persistent, multi-sensor logging made the flight reconstructable.** The 10 Hz on-board record, boot counter, state, reset reason, and validity fields allowed the flight timeline to be recovered even across the reboot.
2. **State persistence is valuable, but power continuity is still a single-point concern.** Restoring Landing after the reset prevented an unsafe state regression; it did not remove the need to identify the physical power interruption.
3. **A validity flag is not the same as measurement plausibility.** The barometric channel remained valid despite its large high-altitude disagreement with GNSS, and isolated battery ADC outliers passed the recorded validity logic.
4. **Receive-side TT&C data need configuration and geometry context.** Range, station motion, antenna attachment, cable configuration, and orientation must be logged alongside RSSI and SNR before decode failures can be attributed to a specific cause.
5. **Payload fault flags identified the affected subsystem but not the mechanism.** Coral timeouts were visible and recovered automatically, yet the available fields cannot distinguish framing, buffering, UART, or payload response delays.
6. **Flight-state names must be checked against mission intent.** The direct Ascent-to-Descent transition was consistent with the implemented rule order, but Cruise was never entered. Requirements and transition priority should agree before the next flight.

15 Future work

Recommended actions before the next flight, in priority order, are:

1. **Investigate the power-on reset.** Inspect battery and regulator logs, connector retention, UVLO thresholds, brownout configuration, reset-cause capture, and the timing relationship between voltage transients and reboot.
2. **Harden the Coral UART path.** Correlate each timeout with receive-state diagnostics, buffer high-water marks, SD activity, and frame duration. Add explicit per-attempt outcome logging so missing sequences can be attributed without inference.
3. **Recalibrate barometric altitude.** Recompute altitude from pressure using a documented reference pressure and atmosphere model; compare residuals against GNSS altitude versus pressure, temperature, and ascent/descent direction.
4. **Validate EPS ADC sampling.** Reproduce switching loads on the bench, inspect reference and divider settling, and add filtering or plausibility rejection for isolated samples outside the electrical design envelope.
5. **Repeat the TT&C link test.** Reattach and tune the main antenna, measure return loss, log ground-station GNSS, and repeat controlled range and orientation steps so antenna and propagation effects can be separated.
6. **Review FSM requirements.** Confirm whether skipping Cruise is acceptable. If Cruise is required, test transition ordering and persistence conditions against the observed apex profile.
7. **Unify the public analysis master.** Restore or explicitly join the 81 GNSS-time-zero rows using boot count and HAL tick so downstream users do not mistake a selection artefact for a logging outage.

16 Conclusions

The public flight record is coherent and supports a defensible reconstruction of pre-flight Standby, launch, ascent, burst/descent, landing, and recovery operations. Core GPS, IMU, barometer, battery, I2C, SD, and LoRa health remained nominal from the flight software's perspective. The most consequential findings are the Coral timeout recurrence, the mid-landing power-on reset, the high-altitude barometric/GNSS disagreement, and the limited interpretability of the recovery link with the main antenna detached. The platform's persistent state and detailed logging worked well; closing the identified power, payload, measurement-validation, and test-traceability gaps should be the priority for the next campaign.

A Reproducibility

Run the following from the repository root on a system with Node.js and TeX Live (including `latexmk`, PGFPlots, and Poppler):

```
powershell -ExecutionPolicy Bypass -File flight-data/report/build_report.ps1
```

The build performs four stages: regenerates the numerical summary and compact CSVs, compiles each standalone vector graph, compiles this report, and copies the final PDF to `output/pdf/Lakitu_Flight_Data_Report.pdf`.

The machine-readable summary is `flight-data/report/analysis_summary.json`. The plot-driving CSVs are in `flight-data/report/processed/`. They are derived artefacts; the original flight-data files remain unchanged.

B Input checksums

Input	SHA-256
Public flight window	0cbe579b5d2ecc1186afbdbc2a8ac285f2cfefe9db8a2e221402e52489cc7ea0
Full archive CSV	ebad4569d5744696ed02636cfecc19c88a1d810e2ea28003309a15cdf56aeb3d
Ground telemetry session 1	e734fff42dad4bd7825452d655f964e40afc276da179c462f246623c29bff3cd
Ground telemetry session 2	c85d80ade6346218438a46a059148746bfbfb9ff5eed71634be2f6826bbf8e31